

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's output space is a discrete label among up to four MCQ options [Q29, Q51], with the 41-subject taxonomy serving as metadata for analysis [Q38]. The deployment targets open-ended Bengali story generation, where the output space is free-form narrative text — fundamentally incompatible with a closed MCQ label space [DATASET-D5 OF concern]. Furthermore, the correct-answer category encodings reflect Indian national/state factual standards — Indian Election Commissioner removal procedure, Indian state political wins, Indian Pharmacopoeia [DATASET-D2, D3, D14] — which encode non-Bangladeshi values and assumptions in the answer-key structure itself. The MCQ output ontology cannot be re-mapped to evaluate culturally grounded narrative generation, and within-Bengali items the answer correctness is calibrated to Indian rather than Bangladeshi knowledge frameworks. No mechanism distinguishes India-specific from Bangladesh-specific answer correctness.

ID	Evidence
Q29	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q51	""For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)
D2	ফেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিানসভা নব্বিবাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি কোন রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D3	নব্বিবাচন কমিশনারকে অপসারণ করা যতে পারে ... প্রধান নব্বিবাচন কমিশনারের সুপারশি রাষ্ট্রপতি — Removal of India's Election Commissioner — refers to Indian constitutional provisions
D5	৬৫তম ফলিমফয়ার অ্যাওয়ার্ডস, ২০২০-এ সরো সনিমোটো গ্ৰাফরি জন্য পুরস্কারটিকে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D14	জাতীয় খাদ্য সুরক্ষা মশিন একট ফিসল উন্নয়ন প্রকল্প ... জাতীয় খাদ্য সুরক্ষা মশিন আগস্ট ২০০৭ সালে চালু হয়েছিলি — India's National Food Security Mission — Indian government program

Strengths

- Structured 4-option categorical output is reproducible and unambiguously scorable [Q51], which provides a clean baseline for measuring factual recall on whatever knowledge frame is encoded — useful as a first-pass screen even if not for the deployment's primary task.

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Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels are India-calibrated correct answers [DATASET-D3 — Indian CEC removal procedure; DATASET-D2 — Indian state election winner]. For a Bangladeshi user, the 'correct' answer for governance items presupposes Indian constitutional structures, not Bangladeshi ones.

OO-2: Missing: any output category encoding Bangladeshi-specific correct answers (e.g., Bangladesh's Election Commission structure, SSC/HSC curriculum standards, Bangladeshi historical events). No story-generation-relevant output categories exist at all.

OO-3: Yes — categories encode Indian national values/assumptions: Indian-state-level governance correctness [DATASET-D2, D3], Indian historical canon [DATASET-D6, D7], Indian cultural figures [DATASET-D4, D5]. These violate convergent validity for a Bangladeshi-targeted evaluation.

OO-4: Stakeholder-driven taxonomy redesign would be required: a story-generation rubric grounded in Bangladeshi cultural touchstones, register, and named entities is needed — MCQ accuracy cannot be re-purposed. BanglaSocialBench's native-speaker-written context-dependent methodology [WEB-16] is the closest existing model.

OO-5: Severe structural and content validity violations: the output taxonomy is doubly misaligned (closed-form MCQ vs. open generation; Indian answer-keys vs. Bangladeshi correctness).

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D2	ফব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিনসভা নির্বাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি কংগন রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D3	নর্বিবাচন কমশিনারকে অপসারণ করা যতে পারে ... প্রধান নর্বিবাচন কমশিনারের সুপারশি রাষ্ট্রপতি — Removal of India's Election Commissioner — refers to Indian constitutional provisions
D4	'মঙ্গল ভারত' কাজটিনমিনলখিতি কংগন জাতীয় নতোর দ্বারা রচিত হযছেলি ... আচার্য বনিবা ভাবে — "Mangal Bharat" — Indian national leader Vinoba Bhave; India-specific cultural/literary content
D5	৬৫তম ফলিমফয়ার অ্যাওয়ার্ডস, ২০২০-এ সরো সনিমোটোগ্রাফরি জন্য পুরস্কারটিকে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D6	_____ দলিলতিে কুতুব মনিারেরে নরিমাণ সম্পন্ন করছেলি ... ইলতুতমশি — Qutub Minar, Delhi — India-specific historical monument
D7	বাহামানি সাম্রাজ্যেরে প্রথম রাজধানী কী ছিলি? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
WEB-15	BLUCK uses Bangladesh-sourced MCQ ontology as a category-redesign reference (https://arxiv.org/abs/2505.21092)
WEB-16	BanglaSocialBench's native-speaker-written, context-dependent evaluation methodology is the closest precedent for a story-generation rubric (https://arxiv.org/abs/2603.15949)

Information Gaps

- Whether any items in the test split have answer keys that would also be correct under Bangladeshi knowledge frameworks (universal STEM items presumably yes; cultural/governance items presumably no)

Requires Expert Verification

- Bangladeshi expert review of cultural/governance answer keys to identify items where Indian and Bangladeshi correct answers diverge

Remediation

Gap 1: Closed MCQ output space cannot evaluate open-ended story generation; correct-answer categories encode Indian rather than Bangladeshi knowledge

Recommendation: Construct a story-generation rubric with native Bangladeshi linguists and literary scholars, modeled on BanglaSocialBench's native-speaker-written context-dependent methodology [WEB-16], covering cultural authenticity, register fidelity, named-entity grounding, and narrative coherence.

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